

Spreading 35 Points in the Plane

Limin Ge*

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Abstract

We place 35 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 32.07769+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 36-point configuration gives 32.83385; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 35 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 32.07769\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 35$ is not on the page. Deleting a point from our 36-point configuration gives 32.83385; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n35.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
32077691908089771053498105
denominator:
999999999998204871013097
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

35 points, x and y per line. File: friedman-maxmin2d/coords/n35.txt.

```
3.846717850121 -3.130720058220
0.904799697733 0.402373578766
4.880797538719 -0.378905743593
2.862898132202 -2.951558782735
3.467213496342 -4.055910009237
2.900652564470 -0.285394824783
6.151526911180 -0.963168029643
4.221223925403 -1.406149606474
1.599345954783 -2.123355125996
4.780511590768 -3.488531810303
4.308459525830 0.441112060483
0.514406484432 -0.518274637986
2.328314551581 0.534622979293
5.545469638594 -2.844451702655
0.902390813233 -2.840469852289
5.993995960677 -1.950682079990
1.871907954406 -3.085493347023
5.242437255276 0.798443280638
4.465990476333 1.428626110830
5.814775268166 -0.021574523438
3.324639807911 0.620273335969
3.574155850499 -2.168581837193
3.482170758414 1.607787386316
1.551867772637 1.164805809485
3.237404207484 -1.226988330988
0.565639170218 -1.898876346084
2.253584489565 -1.047827055502
5.217549181734 -1.320499249798
2.485845502084 1.522137029640
2.483393778423 -3.876748733751
1.681246476676 -0.227809251426
3.884472282389 -0.464556100269
2.590336132580 -1.989420561707
4.599070916494 -2.471867464990
1.262594311769 -1.181761619791
```